WIRELESS TECHNOLOGY For DRIVING THE CARS Of The Future

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The future of the automotive industry lies in connecting vehicles with almost every physical thing in its surrounding and to the cloud

Wireless technologies are revolutionising connectivity in the modern automobile. Ensuring that these technologies operate as expected has become increasingly important.

To give the vehicles a more complete view of the world around, technologies like camera, radar and radio frequency (RF) communication work together with sensor fusion. New vehicles are integrating technologies that help provide advanced driver assistance system (ADAS) functionality, which will eventually lead to fully autonomous operations.

Evolution of wireless technology in automotives

Wireless technologies for automotives advanced with wireless locking and unlocking of doors using infrared (IR). Later, the industry adapted secure and encrypted RF technology for automatic garage door opening and closing with the door lock/unlock mechanism of the automobile.

Then came Bluetooth technology, which increased the convenience and comfort of the users. They could switch on/off appliances, like heaters and air-conditioners, wirelessly. Later, it became a key feature for infotainment systems, apart from providing vehicle diagnostics to the users through connected apps.

Concept of intelligent transport

Use of Bluetooth technology has improved the overall safety of car users. Wireless connections relay high-quality audio, so that passengers can listen to their favourite music (stored on portable devices) and/or hook up their phones to the car.

Wi-Fi provides a robust data pipe for screen projection technologies. It enables seamless transition for the drivers, from their smartphones to cars of their own or rented from anywhere in the world.

Wireless technology applications for connected vehicles

Jeff Phillips, head - automotive marketing, National Instrument, says, "Vehicle-to-everything (V2X) communications require a framework to address the rapidly expanding compliance and certification requirements related to connectivity protocols and sensor control algorithms.

"V2X communications encompass both vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I). We have been working with radar, camera and RF communications for different industries, and natively support all input/output (I/O) types that need to be tested in today's and tomorrow's vehicles."

Srinivasan Subramani, senior technical architect, Mistral Solutions, says, "As design service providers, we provide systems-on-module (SoM) solutions based on Qualcomm 820 and IMX systems-on-chip (SoC), for building infotainment systems. We are also working on 77GHz radars for collision warning, blind-spot detection and reverse assist applications in trucks and cars.

"We follow protocols and standards for design like CAN, LoRA, Bluetooth and Wi-Fi. Our embedded hardware and software solutions integrate audio, video, wireless technologies and DSP algorithms,